SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Dharshavardhan.SB (192420039) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. **Scenario: Event Communication Security**
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. **Scenario: Intercepted Suspicious Message**
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
   a. Deduce the most likely original message.
   b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. **Scenario: Secure Messaging System**
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. **Scenario: Suspicious Digital Asset Investigation**
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. **Scenario: Legacy Encryption in Corporate Communication**
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".
   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

CO1 AT1

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① Scenario : Event communication Security.

· Message: MEET AT NOON

a) Use :- Caesar Cipher shift of +4 (Encrypt)

Org: M E E T A T ~~A F T E R~~ N O O N
+4: Q I I X E X ~~E J X I V~~ R S S R

☺ :- Encrypted Message :- QIIXEXRSSR

b) Decryption : Do 4 position Backward

ie :- Q I I X E X R S S R => M E E T A T N O O N

∴ Recovered the message.

② Scenario intercepted Suspicious Message :

Encrypted :- GSRH RH Z HVXIVG

Atbash cipher :- Alphabet is Reversed A=Z, B=Y...

Org: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Rev: Z Y X W V U T S R Q P O N M L K J I H G F E D C B A

GSRH = THIS
RH = IS
Z = A
HVXIVG = SECRET

∴ ORIGINAL MSG :- "This is a Secret"

(3) Secure Messaging System

Message :- HELLO
Keyword : KEY
Vigenere cipher:.

HELLO
KEYEK

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

HELLO

Msg	key	calc	Ciphers
H	K	7+10 = 17	R
E	E	4+4 = 8	I
L	Y	11+24 = 35 → 9	J
L	E	11+10 = 21	V
O	K	14+4 = 18	S

Therefore :- Encrypted Messag:
HELLO => RIJVS

Scenario :- Suspicious digital Asset investigation
A company logo may contain Hidden confidential information using Stegnalysis technique

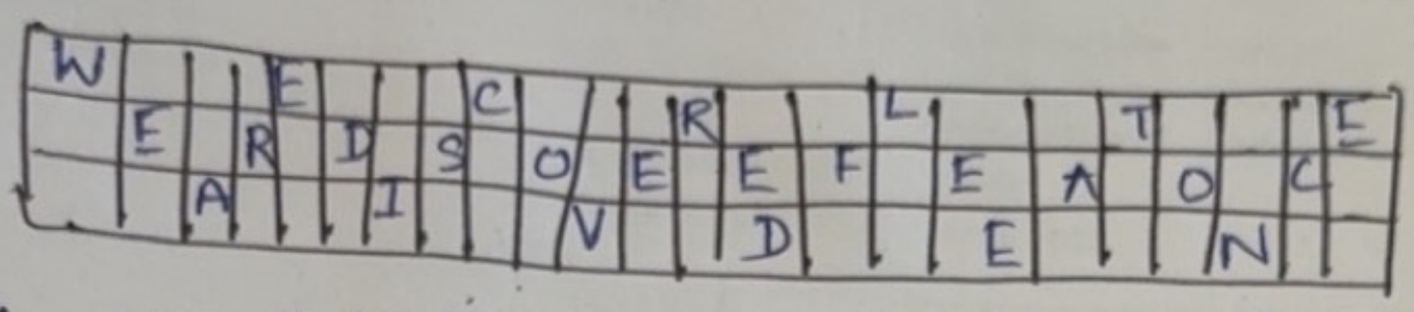
a) How can We detect :-
① : Inspect Image Metadata
② . Analyze pixel Values
③ . Compare file Size
④ . Use Stegnalysis Software
⑤ . Compare With original Image.

b) Two techniques to Strengta Security
①. Encrypt before embedding
②. Use Randomized / adaptive embedding

Scenario : Legacy encryption in corporate communication.

a) Ciphertext : WECRLTEERDSOEEFEAOCAIVDEN

Level 3 - (Railfonce) cipher.



∴ WE ARE DISOVERED FLEE AT ONCE

b) Why Vulnerable : A Transposition cipher is weak in modern communication because :

① . It does not change atchual letters
② . Letter frequency remains unchanged
③ . There are relatively few possible arrangements
④ . attackers can try different Rail arrangements
⑤ . Known words help reveal original message
⑥ . Modern computers easily test Possible arrangement